Faculty of Technology and Engineering

Chandubhai S. Patel Institute of Technology (CSPIT)

Department of Computer Science & Engineering

Date: 19 / 12 / 2025

Laboratory Manual

Academic Year	:	2025-26	Semester	:	4
Course code	:	ITUC202	Course name	:	FUNDAMENTALS OF DATABASE MANAGEMENT SYSTEMS

Practical - 1

Aim: Global Trust Bank is expanding its operations and requires a robust database management system to efficiently manage its employees, job profiles, customers' accounts, and loan information. The bank has laid out specific requirements and constraints to ensure data integrity, uniqueness, and completeness.

Requirements

Employee Management:

- Job Profiles: Maintain records of different job profiles.
- Employees: Store detailed information about employees, including their association with job profiles.

Customer Management:

- Accounts: Maintain separate records for customers' bank accounts.
- Loans: Maintain separate records for customers' loan details
- Design and implement the schema as per the given information.

Constraints –

- Not Null Constraints: Critical fields must not be null to ensure data completeness.
- Unique Constraints: Certain fields must have unique values to avoid duplicates (e.g., Account Number).
- Check Constraints: Enforce domain integrity by limiting the values that can be placed in a column.

1. Create Table Job (job_id, job_title, min_sal, max_sal)

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

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Connections

Oracle Connections
- 24CS021
  - Tables (Filtered)
    - BORROW
      - LOANNO
      - CNAME
      - BNAME
      - AMOUNT
    - DEPOSIT
    - EMPLOYEE
    - JOB
      - JOB_ID
      - JOB_TITLE
      - MIN_SAL
      - MAX_SAL
  - Views
  - Indexes
  - Packages

Reports

All Reports
- About Your Database
- All Objects
- Analytic View Reports
- Application Express
- ASH and AWR
- Database Administration
- Data Dictionary
- Data Dictionary Reports
- Data Modeler Reports
- OLAP Reports
- PLSQL
- Security
- Streams
- Table
- TimesTen Reports
- User Defined Reports
- XML

Welcome Page | 24CS021.sql

Worksheet | Query Builder

```
CREATE TABLE job(
job_id Varchar(15)PRIMARY KEY,
job_title Varchar(30),
min_sal Number(7,2),
max_sal Number(7,2)
);

SELECT * FROM JOB;
DESC JOB;

INSERT INTO Job( job_id , job_title , min_sal , max_sal)
VALUES ('IT_PROG', 'Programmer', 4000, 10000);
INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('MK_MGR', 'Marketing Manager', 9000, 15000);
INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('FI_MGR', 'Finance Manager', 8200, 12000);
INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('FI_ACC', 'Account', 4200, 9000);
INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('LEC', 'Lecturer', 6000, 17000);
INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('COMP_OP', 'Computer Operator', 1500, 13000);

CREATE TABLE Employe(
```

Script Output | Query Result

SQL | All Rows Fetched: 6 in 0.006 seconds

	JOB_ID	JOB_TITLE	MIN_SAL	MAX_SAL
1	IT_PROG	Programmer	4000	10000
2	MK_MGR	Marketing Manager	9000	15000
3	FI_MGR	Finance Manager	8200	12000
4	FI_ACC	Account	4200	9000
5	LEC	Lecturer	6000	17000
6	COMP_OP	Computer Operator	1500	13000

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

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INSERT INTO Job( job_id , job_title , min_sal , max_sal)
 VALUES ('COMP_OP', 'Computer Operator', 1500, 13000);

CREATE TABLE Employe(
```

Script Output | Query Result

Task completed in 0.121 seconds

```
Name       Null?    Type
---------  -------- ------------
JOB_ID     NOT NULL VARCHAR2(15)
JOB_TITLE           VARCHAR2(30)
MIN_SAL             NUMBER(7,2)
MAX_SAL             NUMBER(7,2)
```

2) Create table Employee (emp_no, emp_name, emp_sal, emp_comm, dept_no)

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

```
CREATE TABLE Employee(
emp_no Number(3)PRIMARY KEY,
emp_name Varchar2(30),
emp_sal Number(8,2),
emp_comm Number(6,1),
dept_no Number(3)
);

INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (101, 'Smith', 800, 455, 20);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (102, 'Snehal', 1600, 0, 25);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (103, 'Adama', 1100, 425, 20);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (104, 'Aman', 3000, NULL, 15);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (105, 'Anita', 5000, 50000, 10);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (106, 'Anamika', 2975, NULL, 30);

SELECT * FROM employee;
DESC employee;
```

All Rows Fetched: 6 in 0.007 seconds

	EMP_NO	EMP_NAME	EMP_SAL	EMP_COMM	DEPT_NO
1	101	Smith	800	455	20
2	102	Snehal	1600	0	25
3	103	Adama	1100	425	20
4	104	Aman	3000	(null)	15
5	105	Anita	5000	50000	10
6	106	Anamika	2975	(null)	30

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

```
CREATE TABLE Employee(
emp_no Number(3)PRIMARY KEY,
emp_name Varchar2(30),
emp_sal Number(8,2),
emp_comm Number(6,1),
dept_no Number(3)
);

INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (101, 'Smith', 800, 455, 20);
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 VALUES (102, 'Snehal', 1600, 0, 25);
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 VALUES (103, 'Adama', 1100, 425, 20);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (104, 'Aman', 3000, NULL, 15);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (105, 'Anita', 5000, 50000, 10);
INSERT INTO Employee(emp_no , emp_name , emp_sal , emp_comm , dept_no)
 VALUES (106, 'Anamika', 2975, NULL, 30);

SELECT * FROM employee;
DESC employee;
```

Task completed in 0.49 seconds

```
MAX_SAL                   NUMBER(7,2)
Name      Null?    Type
--------- -------- ------------
EMP_NO    NOT NULL NUMBER(3)
EMP_NAME           VARCHAR2(30)
EMP_SAL            NUMBER(8,2)
EMP_COMM           NUMBER(6,1)
DEPT_NO            NUMBER(3)
```

3) Create table deposit (a_no, name,bname, amount,a_date).

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

```
CREATE TABLE deposit (
a_no Varchar2(5)PRIMARY KEY,
cname Varchar2(15),
bname Varchar2(10),
amount Number(7,2),
a_date Date
);

INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('101', 'Anil', 'Andheri', '7000', '01-jan-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('102', 'Sunil', 'Virar',' 5000', '15-jul-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('103', 'Jay', 'Villeparle',' 6500', '12-mar-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('104', 'Vijay', 'Andheri', '8000', '17-sep-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('105', 'Keyur', 'Dadar',' 7500','19-nov-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('106', 'Mayur', 'Borivali',' 5500', '21-dec-2006');

SELECT * FROM deposit;
DESC deposit;
```

All Rows Fetched: 6 in 0.007 seconds

	A_NO	CNAME	BNAME	AMOUNT	A_DATE
1	101	Anil	Andheri	7000	01-01-06
2	102	Sunil	Virar	5000	15-07-06
3	103	Jay	Villeparle	6500	12-03-06
4	104	Vijay	Andheri	8000	17-09-06
5	105	Keyur	Dadar	7500	19-11-06
6	106	Mayur	Borivali	5500	21-12-06

Oracle SQL Developer : C:\Users\URJA GAJERA\24cs021 DBMS\24CS021.sql

```
CREATE TABLE deposit (
a_no Varchar2(5)PRIMARY KEY,
cname Varchar2(15),
bname Varchar2(10),
amount Number(7,2),
a_date Date
);

INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('101', 'Anil', 'Andheri', '7000', '01-jan-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('102', 'Sunil', 'Virar',' 5000', '15-jul-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('103', 'Jay', 'Villeparle',' 6500', '12-mar-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
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INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('105', 'Keyur', 'Dadar',' 7500','19-nov-2006');
INSERT INTO Deposit(a_no , cname , bname , amount , a_date)
VALUES ('106', 'Mayur', 'Borivali',' 5500', '21-dec-2006');

SELECT * FROM deposit;
DESC deposit;
```

Task completed in 0.466 seconds

```
DEPT_NO           NUMBER(3)
Name    Null?     Type
------ -------- ------------
A_NO    NOT NULL VARCHAR2(5)
CNAME            VARCHAR2(15)
BNAME            VARCHAR2(10)
AMOUNT           NUMBER(7,2)
A_DATE           DATE
```

4)Create table borrow (loan no,cname,bname,amount).

Oracle SQL Developer

```
CREATE TABLE borrow(
loanno Varchar2(5)PRIMARY KEY,
cname Varchar2(15),
bname Varchar2(20),
amount NUMBER(7,2)
);

INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('201', 'ANIL', 'VRCE', '1000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('206', 'MEHUL', 'AJNI', '5000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('311', 'SUNIL', 'DHARAMPETH', '3000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('321', 'MADHURI', 'ANDHERI', '2000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('375', 'PRMOD', 'VIRAR', '8000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('481', 'KRANTI', 'NEHRU PLACE', '3000.00');

SELECT * FROM BorroW;
DESC Borrow;
```

Task completed in 0.221 seconds

```
LOANN CNAME           BNAME                    AMOUNT
----- --------------- -------------------- ----------
201   ANIL            VRCE                       1000
206   MEHUL           AJNI                       5000
311   SUNIL           DHARAMPETH                 3000
321   MADHURI         ANDHERI                    2000
375   PRMOD           VIRAR                      8000
481   KRANTI          NEHRU PLACE                3000
```

Oracle SQL Developer : C:\Users\Administrator\24CS021~1.sql

```
CREATE TABLE borrow(
loanno Varchar2(5)PRIMARY KEY,
cname Varchar2(15),
bname Varchar2(20),
amount NUMBER(7,2)
);

INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('201', 'ANIL', 'VRCE', '1000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('206', 'MEHUL', 'AJNI', '5000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('311', 'SUNIL', 'DHARAMPETH', '3000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('321', 'MADHURI', 'ANDHERI', '2000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('375', 'PRMOD', 'VIRAR', '8000.00');
INSERT INTO Borrow(loanno , cname , bname , amount)
VALUES ('481', 'KRANTI', 'NEHRU PLACE', '3000.00');

SELECT * FROM BorroW;
DESC Borrow;
```

Task completed in 0.221 seconds

```
481   KRANTI          NEHRU PLACE                3000

6 rows selected.

Name   Null?    Type
------ -------- ------------
LOANNO NOT NULL VARCHAR2(5)
CNAME           VARCHAR2(15)
BNAME           VARCHAR2(20)
AMOUNT          NUMBER(7,2)
```